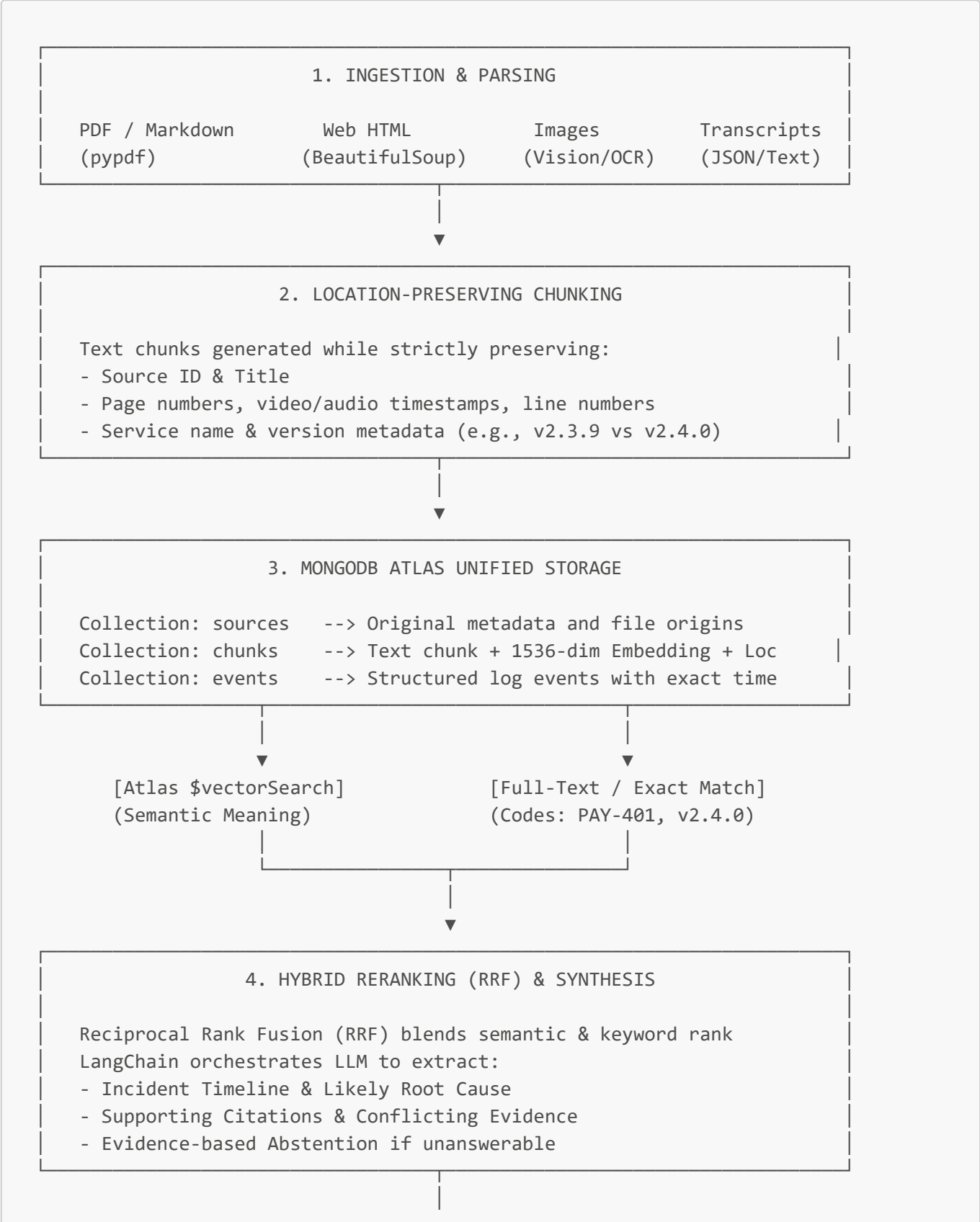
Chapter 2: Technical Architecture & Deep Dive

1. High-Level System Architecture

TraceIQ is built as a modular system with four core stages:



```
                                    ▼
┌──────────────────────────────────────────────────────────────────────┐
│                    5. NEXT.JS INVESTIGATOR UI                        │
│                                                                      │
│   Interactive Incident Timeline  |  Evidence Graph                   │
│   Source Inspector Drawer        |  One-Click Citation Jumper        │
└──────────────────────────────────────────────────────────────────────┘
```

---

2. What is an Embedding?

Computers cannot inherently understand words; they only understand numbers.

An **Embedding Model** (such as OpenAI's `text-embedding-3-small`) reads a chunk of text and translates its semantic meaning into a list of numbers (a vector), typically 1,536 numbers long:

```
# Example representation of an embedding vector:
[0.0124, -0.0451, 0.8291, ..., 0.0039]  # 1536 numbers
```

- Phrases with similar meanings end up close together in mathematical space:
  - "*payment gateway failure*" and "*checkout HTTP 500 error*" have high similarity (~0.89).
  - "*lunch menu cafeteria*" and "*database connection pool*" have very low similarity (~0.12).

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3. MongoDB Vector Search vs Keyword Search

Neither Vector Search nor Keyword Search is enough on its own:

Why Vector Search Alone Isn't Enough

- **Strength**: Understands synonyms and concepts. Searching "*why did payments break*" will find "*credit card processing rejected*".
- **Weakness**: Completely blind to exact alphanumeric codes. If you search for error code `PAY-401`, a vector search might return `ERR-502` because both are "error codes".

Why Keyword Search Alone Isn't Enough

- **Strength**: Finds exact tokens (`PAY-401`, `v2.4.0`, `192.168.1.1`).
- **Weakness**: Completely misses conceptual matches if the exact keyword isn't present in the document.

The Solution: Hybrid Search with Reciprocal Rank Fusion (RRF)

TraceIQ runs **both** searches simultaneously in MongoDB Atlas:

1. Vector Search returns Top-10 candidates based on conceptual similarity.
2. Exact Text Search returns Top-10 candidates matching exact codes.
3. **Reciprocal Rank Fusion (RRF)** combines their ranks using the formula: $$\text{RRF Score} = \frac{1}{60 + \text{Rank}_{\text{vector}}} + \frac{1}{60 + \text{Rank}_{\text{keyword}}}$$ This guarantees that results

matching *both* meaning and exact identifiers appear right at the top.

---

4. Why We Chose This Tech Stack

- **FastAPI (Python)**: The industry standard for AI backends. Python has native support for LangChain, NumPy, PyPDF, and ML libraries, while FastAPI provides high-speed asynchronous endpoints with automatic OpenAPI documentation (`/docs`).
- **LangChain**: A framework that manages prompts, document splitters, and output parsers, keeping your business logic clean and modular.
- **MongoDB Atlas**: Unifies original documents, structured log events, chunks, and vector embeddings in a single database. You don't need a separate vector database (like Pinecone) plus a relational database — MongoDB handles both.
- **Next.js (React + TypeScript)**: Delivers a responsive, server-rendered incident investigation console with rich components for timelines and citation drawers.